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School of Information Technology and Electrical Engineering

EXAMINATION

Semester Two Final Examinations, 2021

ELEC7309 Power System Planning and Reliability

This paper is for St Lucia Campus students.

Examination Duration: 120 minutes

Reading Time: 10 minutes

Exam Conditions:

This is a Closed Book examination - no written materials permitted

Casio FX82 series or a calculator on the UQ approved list

During reading time (= planning time) - students are encouraged to review and plan responses to the exam questions

This examination paper will be released to the Library

Materials Permitted In The Exam Venue:
(No electronic aids are permitted e.g. laptops, phones)
None

Materials to Be Supplied To Students:

2 x 14-Page Answer Booklet

Instructions to Students: There are two Sections in this exam (PART-A and PART-B). Total Marks in the exam is 60%. Each Section is 30%. Answer each section in a separate Booklet.

Additional exam materials (e.g. answer booklets, rough paper) will be provided upon request.

For Examiner Use Only

Question	Mark

Total _____

PART A: POWER SYSTEM RELIABILITY
ANSWER ANY THREE QUESTIONS FROM PART-A
All Questions are of equal value

Question A1 (Total Marks 10)

A power system contains three identical 50 MW generating units each with a 5% FOR and one 100 MW unit with a 4% FOR. The system peak load for a specified 100-day period is 200 MW.

(a) Calculate the LOLE and the expected load loss for this system. Assume that the appropriate load characteristic is a straight line from 100% to 70% points. **(4 Marks)**

(b) The economic loading order for this system is the 100 MW unit first, followed by three 50 MW units (with no order). Calculate the expected energy provided to the system by the 100 MW unit. Assume that the load duration curve for this period is a straight line from the 100% to 75% load points. Calculate the energy index of reliability (EIR) for this system. **(6 Marks)**

Question A2 (Total Marks 10)

Two systems are interconnected by two 15 MW tie lines. System A has four 30 MW generating units with forced outage rates of 5%. System B has eight 15 MW generating units with forced outage rates of 6%. Assume that the tie lines have failure rates of 5 failures/year and average repair times of 24 hours.

(a) Calculate the expected loss of load in each system in days and in MW for a one-day period, given that the peak load in both System A and System B is 100 MW. **(5 Marks)**

(b) What is the LOLE for System A, assuming that the maximum assistance from the system B is fixed at 20 MW, i.e. the peak load reserve margin? **(5 Marks)**

Question A3 (Total Marks 10)

Consider a single line representation of a radial three-phase power system as shown in Figure 1. Assume that any fault, single phase or otherwise, will trip all three phases. It is normally found in practice that lines and cables have a failure rate, which is approximately proportional to their length. For this example, let us assume that the main feeder (sections 1, 2, 3, and 4) have a failure rate of 0.25 f/km/yr and the lateral distributors (a, b, c, and d) have a failure rate of 0.4 f/km/yr. Assume that the repair time *r* for each of the main feeder is 2.5 hours and each of the lateral distributor is 3 hours.

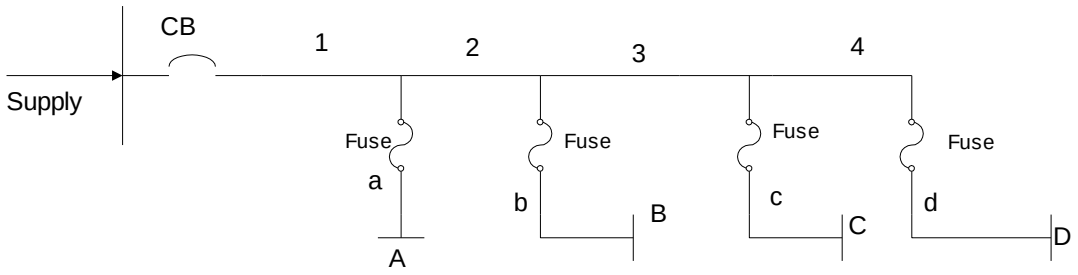


Figure 1 for Question A3

Reliability parameters for the system of Figure 1

Component	Length (km)
Section- 1	5
Section- 2	2
Section- 3	4
Section- 4	2
Distributor -a	1
Distributor -b	1.5
Distributor -c	1
Distributor -d	1.5

Customers and load connected to System in Figure 1

Load point	Number of customers	Average load connected (kW)
A	1100	4000
B	800	3000
C	750	3500
D	550	2000

There is only one main CB. Assume that all component failures are short circuits, and then each failure on the main feeder will cause the main breaker to operate. Let us now install a fuse at the tee-point in each lateral distributor. In this case a short circuit on a lateral distributor causes its appropriate fuse to blow; this causes disconnection of its load point until the failure is repaired but does not affect or cause the disconnection of any other load points. On this basis calculate distribution system indices SAIFI, SAIDI, CAIDI, ASUI and ENS for this system.

Question A4 (Total Marks 10)

Consider the power system shown in Figure 2.

Power System data

Generating stations

1. 4 x 25 MW units λ = 2.0 f/yr μ = 98.0 r/yr
2. 2 x 40 MW units λ = 3.0 f/yr; μ = 57.0 r/yr

Loads
A-85MW
B-65MW

Transmission lines
1. λ = 4f/yr, r = 8 hrs
Load Carrying Capability (LCC) = 85 MW

2. λ = 5 f/yr, r = 8 hrs, LCC = 65 MW

3. λ = 3 f/yr, r = 12 hrs, LCC = 50 MW

Calculate an appropriate set of indices at load points A and B and for the whole system.

Assume that the loads are constant at the values shown, for a one-year period and that the transmission loss in each of the line is zero.

In calculating the composite system adequacy indices, consider up to two simultaneous outages and assume that all load deficiencies are shared equally whenever possible.

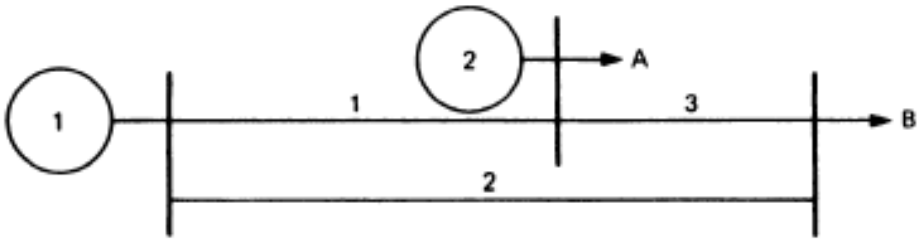


Figure 2 for Question A4

PART B: POWER SYSTEM PLANNING
ANSWER All THREE QUESTIONS FROM PART-B
All Questions are of equal value

Question B1 **(Total Marks 10)**

Net Present Value (NPV) will be calculated for two options, shown in Table 1, utilising a discounted cash flow (DCF) technique to identify the most economical option. Details of the expenditures and benefits for each option are presented in Table 1 below.

Table 1 for Question B1

Options	Costs		Benefits	
	Capital Cost ($)	Operation & Maintenance ($)	Annual System Loss Reduction (MWh)	Annual Expected Energy Not Supplied (EENS) Reduction (MWh)
Option X	4,000,000	175,000	3,500	25
Option Y	7,000,000	190,000	4,000	30

The following data should be used in constructing the DCF:

- Discount rate 4% per annum
- Asset life for both options 7 years
- Average marginal energy cost $28/MWh
- Value of Customer Reliability (VCR) $34/kWh

(a) Perform a DCF for both options using the format shown in Table 2. You can draw Table 2 twice on your answer paper for submission, one for Option X and one for Option Y. **(5 Marks)**

(b) Determine which option/project should be recommended (if any). (**1 Mark)**

(c) Determine what value of VCR would result in Option X becoming break-even. **(4 Marks)**

Table 2 for Question B1

Option: _____

Year	Disc Factor	Costs		Benefits		Annual Cash Flow	DCF
		Capex	Opex	Loss Saving	Reliability		
0							
1							
2							
3							
4							
5							
6							
7							
						NPV:	

Question B2 (Total Marks 10)

Consider two systems depicted in Figure 1. The marginal costs of generation in both systems were governed by the following two equations (with P_X and P_Y in MW).

$$MC_X = 18 + 0.03 \cdot P_X \left[\frac{\$}{\text{MWh}}\right]$$

$$MC_Y = 16 + 0.02 \cdot P_Y \left[\frac{\$}{\text{MWh}}\right]$$

(a) Assume that the transfer capacity between both systems is 0 MW, for this condition calculate the price of electricity and the electricity cost per hour for the customer in each system; calculate the cost of generation production per hour, the revenue per hour of the generator, and the profit made by the generator per hour in each system. **(5 Marks)**

(b) Consider both systems in Figure 1 with the same marginal cost functions, but now the two systems are interconnected without a limit on transfer capacity. Calculate again the electricity price and the electricity cost per hour for the customers in each system; calculate the cost of generation production per hour, the revenue per hour to the generator, and the profit made by the generator per hour in each system. **(5 Marks)**

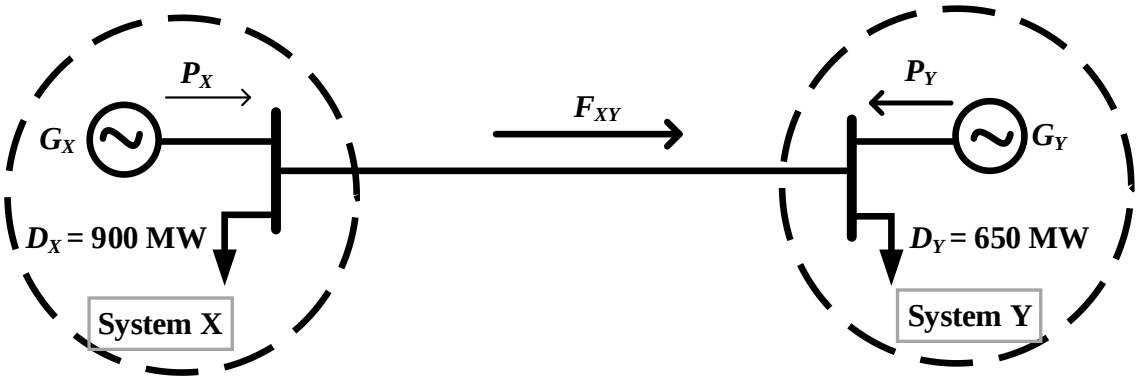


Figure 1 for Question B2: System Details

Question B3 **(Total Marks 10)**

For three generation facilities, X, Y and Z, with details given in Table 3, assume an interest rate of 5% and the possibility of choosing any generation capacity of the above-mentioned technologies (X, Y and Z).

(a) Derive the equations of annual cost as a function of capacity factor for each of the technologies, using the data provided in Table 3 and the Load Duration Curve shown in Figure 2. **(3 Marks)**

(b) Using the Screening Curve technique, determine the optimal type and capacity to supply the demand characterised by the Load Duration Curve shown in Figure 2. **(5 Marks)**

(c) Based on the results obtained in b), calculate the annual operation cost for the most operated generation technology type. **(2 Marks)**

Table 3 for Question B3: Generation technology data

Technology type	Investment cost ($/kW)	Plant life (years)	Fuel cost ($/MWh)
X	1200	35	15
Y	400	28	24
Z	200	24	29

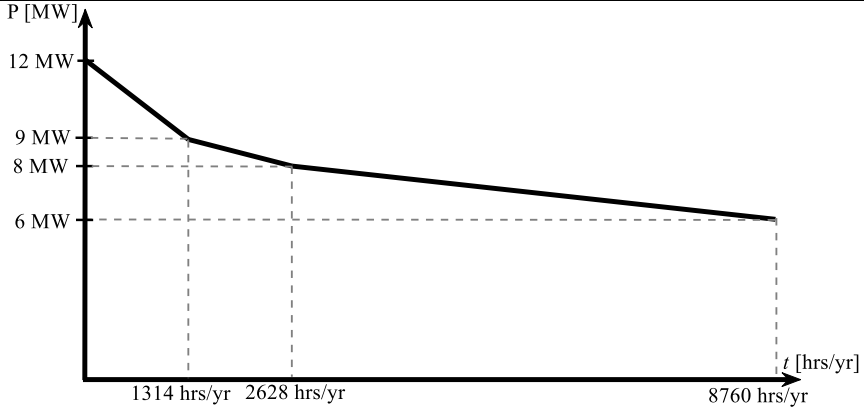


Figure 2 for Question B3: Load Duration Curve

END OF EXAMINATION

Formulae-PART A

General

Failure rate/Forced Outage Rate: λ

Repair Rate: μ

Mean Time Before Failure (MTBF) $\frac{1}{\lambda}$

Mean Time To Repair (MTTR): $r = \frac{1}{\mu}$

For Generation and Composite Power System Reliability Assessment

Availability: $A = \frac{\mu}{\mu+\lambda}$

Unavailability: $U = \frac{\lambda}{\mu+\lambda}$

For Distribution Reliability Assessment

Availability: $A = 1$

Unavailability: $B = \lambda \cdot r = \frac{\lambda}{\mu}$

Common Customer Reliability Indices

System Average Interruption Frequency Index

$$\text{SAIFI} = \frac{\text{total number of customer interruptions}}{\text{total number of customers served}} = \frac{\sum N_i \cdot F_i}{\sum N_i}$$

System Average Interruption Duration Index

$$\text{SAIDI} = \frac{\text{sum of all customer interruption duration}}{\text{total number of customers served}} = \frac{\sum N_i \cdot Q_i}{\sum N_i}$$

Customer Average Interruption Frequency Index

$$\text{CAIFI} = \frac{\text{total number of customer interruptions}}{\text{total number of customers affected}} = \frac{\sum N_i \cdot F_i}{\sum N_i}$$

Customer Average Interruption Duration Index

$$\text{CAIDI} = \frac{\text{sum of all customer interruption duration}}{\text{total number of customer interruptions}} = \frac{\sum N_i \cdot Q_i}{\sum N_i \cdot F_i}$$

Average Service Availability Index

$$\text{ASAI} = \frac{\text{customer hours of available service}}{\text{customer hours demanded}} = \frac{\sum N_i \cdot (1 - Q_i)}{\sum N_i}$$

Average Service Unavailability Index $\text{ASUI} = 1 - \text{ASAI}$

Energy Not Supplied

$$\text{ENS} = \sum P_i \cdot Q_i \cdot 8760 \frac{\text{hrs}}{\text{yr}}$$

Average Energy Not Supplied

$$\text{AENS} = \frac{\text{total energy not supplied}}{\text{numbers of customers served}} = \frac{\sum P_i \cdot Q_i \cdot 8760 \frac{\text{hrs}}{\text{yr}}}{\sum N_i}$$

Average Curtailment Index

$$\text{ACCI} = \frac{\text{total energy not supplied}}{\text{number of customers affected}} = \frac{\sum P_i \cdot Q_i \cdot 8760 \frac{\text{hrs}}{\text{yr}}}{\sum N_i}$$

Composite System Reliability Analysis

If the occurrence of an event A is dependent upon a number of event B_i which are mutually exclusive, then the system failure event A will be;

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i) P(B_i)$$

$$P_k = \sum_j P(B_j)(P_{gj} + P_{lj} - P_{gj}P_{lj})$$

$$F_k = \sum_j [F(B_j)(P_{gj} + P_{lj} - P_{gj}P_{lj})$$

Here;
B_j= outage state j in the transmission network
P_{gj} = probability of generating capacity outage at state j exceeding the reserve margin
P_{lj} = probability of state j at load bus k exceeding the maximum load that can be supplied at that bus without failure.
P_c = probability of transmission inadequacy
Where $F(B_j)$ is the frequency of occurrence of outage B_j

$$F_C = \sum_{j=1}^{2^m} \sum_{i=1;i\neq j}^{2^m} \underbrace{(1 - P_C(j))} \cdot \underbrace{P_C(i)} \cdot \lambda_{ji} \qquad P_C = \sum_{j=1}^{2^m} P(j) \cdot P_C(j)$$

$$F = \underbrace{(1 - P_G)} \cdot F_C + \underbrace{(1 - P_C)} \cdot F_G$$

Probability that generation is sufficient

Probability that network is sufficient

Formulae-PART B

Variable names and meaning

P_0: present value
F_n: value in n years
A: annuity (constant)
A_n: annuity in year n
i: interest rate
B_{jt}: profit of plan j in year t
C_{jt}: operation cost of plan j in year t
K_{jt}: investment of plan j in year t
j: number of the plan in consideration
n_j: horizon of study (service/lifetime) of plan j

Present value

$$P_0 = \frac{F_n}{(1+i)^n}$$

Present Value Factor $(P|F,i,n)$:

$$(P|F,i,n) = \frac{1}{(1+i)^n}$$

Future Value

$$F_n = (1+i)^n \cdot P_0$$

Future Value Factor $(F|P,i,n)$:

$$(F|P,i,n) = (1+i)^n$$

Constant Annuity

$$F_n = A\frac{(1+i)^n - 1}{i} = A(F|A,i,n)$$

$$A = F_n\frac{i}{(1+i)^n - 1} = F_n(A|F,i,n)$$

$$P_0 = \frac{F_n}{(1+i)^n} = A\frac{(1+i)^n - 1}{i\cdot(1+i)^n} = A\frac{1-(1+i)^{-n}}{i} = A(P|A,i,n)$$

$$A = P_0\frac{i\cdot(1+i)^n}{(1+i)^n - 1} = P_0\frac{i}{1-(1+i)^{-n}} = P_0(A|P,i,n)$$

Capital Recovery Factor

$$CRF_j = \frac{i\cdot(1+i)^{n_j}}{(1+i)^{n_j} - 1} = \frac{i}{1-(1+i)^{-n_j}} = (A|P,i,n)$$

Incremental Annuity

$$P_0 = \sum_{k=1}^{n} A_k(1+i)^{-k} = \sum_{k=1}^{n} A_k(P|F,i,k)$$

Net present value

$$NPV_j = \sum_{t=1}^{n_j}\left[B_{jt}(P|F,i,t)\right] - \sum_{t=1}^{n_j}\left[\left(C_{jt}+K_{jt}\right)(P|F,i,t)\right]$$

Benefit/Cost Ratio

$$BCR_j = \frac{\sum_{t=1}^{n_j}\left[\left(B_{jt}-C_{jt}\right)(P|F,i,t)\right]}{\sum_{t=1}^{n_j}\left[K_{jt}(P|F,i,t)\right]}$$

Present Value of Costs

$$PVC_j = \sum_{t=1}^{n_j}\left[\left(C_{jt}+K_{jt}\right)(P|F,i,t)\right]$$